



**Generation Interconnection
System Impact Study Report
for**

Queue Project AE1-051

EAST CARBONDALE-LACKAWANNA 69KV

10 MW Capacities / 0 MW Energy

August, 2019

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

2 General

The IC has proposed an uprate to an existing wind generating facility located Wayne County, Pennsylvania. The AE1-051 represents a request for an increase to the previous Z2-107/K-21/D-05 projects, which all share the same property and connection point. The Z2-107/K-21/D-05 projects have a capability of 74.5 MW with 9.0 MW being recognized as Capacity.

AE1-051 will maintain capability of 74.5 MW (no change from previous queue positions Z2-107/K21/D05) with a requested uprate of 10 MWC of additional output being recognized by PJM as Capacity.

The total capability of the previous projects and AE1-051 uprate will be 74.5 MW with 19.0 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is September 01, 2020.

This study does not imply a PPL EU commitment to this in-service date.

The project capability is summarized below:

	Maximum Facility Output (MW)	Capacity Interconnection Rights (MW)
Existing Z2-107/K-21/D-05	74.5	9
Incremental AE1-051 MW	0	10
Total MW	74.5	19

Queue Number	AE1-051
Project Name	EAST CARBONDALE-LACKAWANNA 69KV
Interconnection Customer	Waymart Wind Farm LLC
State	Pennsylvania
County	Wayne
Transmission Owner	PPL
MFO	74.5
MWE	0
MWC	10
Fuel	Storage
Basecase Study Year	2022

2.1 Point of Interconnection (POI)

AE1-051 interconnects with the PPL EU transmission system via the East Carbondale – Lackawanna 69kV line. AE1-051 will utilize the same POI of the prior Z2-107, K-21 and D-05 queues.

The scope of work provided in this Impact Study is for facilities to be constructed on the PPL EU side of the POI.

2.2 Cost Summary

The AE1-051 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$40,000
Cost Allocation for System Upgrades	\$0*
Total Costs	\$40,000

* Note: Stability analysis and evaluation of NYISO impacts will be performed during the Facilities Study phase. Additional upgrades may be identified as part of those studies.

2.3 Estimated Schedule

The estimated time to complete the scope of work is **4-6 months** after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

2.4 Assumptions

- Availability of optimal transmission line route assumptions are not applicable for this update
- Outage feasibility not assessed until Facilities Study
- No major environmental, real estate, or permitting issues

3 Transmission Owner Scope of Work

3.1 Attachment Facilities

None.

3.2 Direct Connection Cost Estimate

None.

3.3 Non-Direct Connection Cost Estimate

- Conduct a full relay settings analysis to determine sufficient system protection
 - Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities.
 - Conduct a review of the existing PPL IC IPR, POC protective relays settings and IC's engineering package (submitted by IC to PPL EU)
 - Identify affected relays
 - Revise settings as needed
 - Implement revised settings as needed
 - Perform system tests and checks to ensure revised settings functionality

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Relay Settings Analysis, CAPE Revisions and Short Circuit Study	\$40,000
Total Non-Direct Connection Facility Costs	\$40,000

4 Interconnection Customer Requirements

4.1 PPL EU Direct Transfer Trip (DTT) Requirements

PPL EU will require an independent communication path, for DTT of the IC Intertie Protective Relaying (IPR) Fault Interrupting Devices (FIDs), consisting of one communication circuit with. To ensure reliable communication, the IC shall also provide DTT relaying equipment identical to the PPL EU DTT relaying equipment. All DTT relaying equipment shall connect to the respective communication path. All DTT relaying equipment should reside within the same location as the IPR and Point of Contact (POC) relaying equipment.

PPL EU assumes existing communication path will continue to be used for the independent communication pathway.

5 Revenue Metering and SCADA Requirements

5.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

5.2 PPL Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

6 Network Impacts

The Queue Project AE1-051 was evaluated as a (Capacity 10.0 MW) injection at **East Carbondale-Lackawanna 69kV** in the PPL area. Project AE1-051 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-051 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

7 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

8 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

9 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

10 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

11 Stability and Reactive Power Requirement for Low Voltage Ride Through

Stability Analysis will be performed during the Facilities Study. **Note that additional reinforcements may be identified to mitigate stability concerns.**

12 Light Load Analysis

No mitigations required.

13 System Reinforcements

None

Affected Systems

14 Affected Systems

14.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

Short Circuit

15 Short Circuit

The following Breakers are over duty

None

Attachment 1

Single Line Diagram

